UTM-YOLOX: Unknown Target Mining-Based Open-Set SAR Ship Detection and Classification

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***Abstract*—Existing synthetic aperture radar (SAR) ship detection and classification methods based on deep learning have achieved notable performance under the close-set assumptions. However, SAR object detection and classification tasks inherently operate in open-set scenarios, which naturally involve the potential presence of unknown noncooperative targets. Closed-set methods cannot identify these unknown targets, often misclassifying them as known classes or background, leading to significant errors. Therefore, this letter proposes an unknown-target mining-based open-set SAR ship detection and classification method UTM-YOLOX, which preserves the performance for known targets while enhancing the ability to detect and identify unknown targets. First, class-agnostic localization quality estimation (CLQE) is utilized in the regression heads of YOLOX to improve the detection of unknown targets. Unlike binary foreground-background classification, CLQE focuses on learning the quality of target localization, providing richer information for target detection. Second, an open-set classifier (OSC) is proposed for classification heads of YOLOX to identify the unknown targets and known classes, which includes unknown class discriminant learning (UCDL) and contrastive clustering learning (C2L). UCDL utilizes the detected proposals of suspected unknown targets into the training of the OSC, enabling the model to effectively identify unknown targets, while C2L is utilized to improve the discriminative ability between known and unknown targets. Experiments on SRSDD-v1.0 demonstrate that UTM-YOLOX achieves satisfactory performance. The code is available at https://github.com/LiYiming98/UTM-YOLOX**

***Index Terms*—Open-set classifier (OSC), open-set learning, synthetic aperture radar (SAR) ship detection and classification, YOLOX.**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active remote sensing technology capable of high-resolution microwave imaging, offering advantages like all-weather operation, day-and-night functionality. SAR can perform tasks that other remote sensing methods find challenging, making it highly valuable in both military and civilian applications. SAR image interpretation forms the foundation for these applications, with SAR target detection and classification being the key focus. By detecting and classifying targets in large-scale SAR images, accurate positioning and identification of these targets can be achieved, providing significant practical value.

With the continuous improvement of SAR technology, high-resolution SAR images are becoming more accessible, supporting the application of deep learning for target detection and classification. Unlike multistage processing in traditional SAR target detection and classification, deep learning-based algorithms streamline detection and classification into a unified process, reducing system complexity and cost. As a result, this integrated approach has become the mainstream trend in SAR target detection and classification research.

Most deep learning-based target detection algorithms are developed under the closed-set assumption, which assumes that the target classes in both training and testing are the same. However, in real world, there are targets from new classes not seen or not labeled in the training dataset. We call these targets unknown or out of distribution (OOD) [1]. Models trained under the closed-set assumption are prone to misclassifying unknown targets as background or misclassifying them as known classes, leading to a decrease in performance, which is known as *open-set risks* [1]. In the context of SAR ship detection and classification, this problem is even more pronounced. The complex and cluttered nature of SAR images makes it difficult to distinguish unknown targets from background clutter, further exacerbating the challenge.

To mitigate the open-set risks, open-set detection and classification has gained widespread attention in the optical images. The open-set detector trained on closed-set dataset needs to detect and identify unknown targets as "unknown." The presence of unknown targets in training set ($\mathrm{OOD}_{\mathrm{unlabeled}}$) [2], [3] and their absence [4] are two distinct assumptions in open-set research. Specifically, $\mathrm{OOD}_{\mathrm{unlabeled}}$ is the prevailing assumption in current studies. For instance, Wang et al. [2] proposed Random box and Zhao et al. [3] proposed PAD-CEC. The core idea of these methods is primarily based on two-stage detectors, which first mine suspected target slices from the background in the training data and then incorporate them as unknown targets for learning in the open-set classification subnetworks. However, unlike optical natural images, SAR target open-set detection and classification methods must address unique challenges specific to the nature of the task as follows.

1) SAR images contain substantial background clutter, posing challenges in reliably discriminating it from unknown targets of interest [5]. Such a challenge is rarely considered in open-set detection and classification methods for optical

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natural images, since they focus on detecting all meaningful objects rather than predefined objects of interest.

2) The attitude sensitivity of radar targets' echoes leads to insufficient interclass feature separability [6], making it difficult for the model to accurately distinguish between known and unknown targets. This challenge also exists in SAR ship detection and classification tasks.

3) Practical applications impose real-time inference requirements [7], which two-stage detectors often struggle to meet compared to one-stage detectors.

In response to these challenges, this letter proposes an open-set SAR ship detection and classification method (UTM-YOLOX) based on unknown target mining, building upon the single-stage detector YOLOX [8]. First, to detect more potential unknown targets, a class-agnostic localization quality estimation (CLQE) is introduced. By learning the localization quality of targets rather than a binary foreground-background classification, the detection ability for known targets can be generalized to unknown targets since targets have more similar localization information compared to the distinction between targets and clutter. Next, to extend the ability to identify known classes, unknown targets, and background clutter, an open-set classifier (OSC) is proposed, which includes the unknown class discriminant learning (UCDL) and contrastive clustering learning (C2L). UCDL uses an uncertainty estimation method to select hard-to-classify targets and mines suspected unknown targets from the background based on localization quality scores to train the OSC. C2L is utilized to increase the compactness of known targets, which enhance the OSC's ability to identify unknown targets. The novelty of our letter can be summarized as follows.

1) We propose an open-set detector for the integrated detection and classification tasks. To the best of our knowledge, it is the first open-set learning technique for SAR ship detection and classification tasks.

2) We innovatively propose an OSC to classify known targets and identify unknown targets of interest from substantial background clutter. The UCDL is designed to mine potential unknown targets of interest from the detected candidate regions to train the OSC together with labeled known targets, and C2L is adopted to enhance the discrimination between known and unknown targets.

3) We adopt the single-stage YOLOX detector [8] as the backbone and design a simple yet efficient open-set detection and classification framework, which significantly reduces computational complexity compared to existing methods [2], [3], [12].

II. Methodology

A. Preliminary

Given a training dataset $D_{\text{tr}} = \{(x, y), x \in X, y \in Y\}$, where x is the input, $y = \{(c_i, b_i)\}_{i=1}^{N}$ is the class labels and the coordinates of the bounding box. D_{tr} contains K known classes $C_K = \{1, \ldots, K\}$. The open-set detection and classification aims to train a detector on D_{tr} that can not only detect and classify targets belonging to C_K but also detect and identify targets belonging to C_U.Open-set detector only needs to identify unknown targets as "unknown," rather than classifying unknown targets into different target classes [8].

B. Overall Framework

The diagram of UTM-YOLOX is shown in Fig. 1. Building upon the single-stage YOLOX detector, this letter introduces: 1) CLQE method in the regression heads, designs and 2) OSC method in the classification heads. We will provide a detailed introduction to three modules.

C. Class-Agnostic Localization Quality Estimation

YOLOX detector utilizes a confidence branch for foreground-background classification learning. However, such a branch faces the risk of overfitting to known targets and misclassifying unlabeled unknown targets as background, which greatly limits unknown targets detection performance.

Inspired by Ge et al. [8], this letter introduces class-agnostic intersection over union (IoU) branch and centeredness branch into the regression heads of YOLOX. These two branches enable the regression heads to focus on the information necessary for target localization. Compared to foreground–background binary classification, this approach can generalize the ability to detect known targets to unknown targets (e.g., different types of unseen ships), while avoiding the risk of misclassifying unlabeled unknown targets as background.

Specifically, CLQE computes the IoU between predicted boxes and ground-truth boxes, and selects those with higher IoU as positive samples [11] to train the class-agnostic IoU and center branches. The training loss is given

$$L_{\text{CLQE}} = \frac{1}{N}\left(\sum_{i=0}^{N}\left|p_i - \text{centerness}_i\right| + \sum_{i=0}^{N}\left|q_i - \text{IoU}_i\right|\right) \tag{1}$$

where p_i is the output of the ith predicted box in the centeredness branch, q_i is the output of the predicted box in class-agnostic IoU branch, and N is the number of positive samples. The calculation method of centeredness can be referred to [9]. In the test stage, CLQE computes the localization quality score for predicted boxes as

$$s = \sqrt{\text{centerness} \times \text{IoU}}. \tag{2}$$

D. Open-Set Classifier

CLQE helps better detect suspected unknown targets. To further distinguish between known targets, unknown targets, and clutter, this letter proposes OSC, which includes UCDL and C2L. First, OSC augments the K-way closed-set classifier with the $K+1$-way OSC, where $K+1$ denotes "unknown." To improve the reliability of mining unknown targets from the background, UCDL is established. While C2L is utilized to improve the discriminative ability between known and unknown targets.

1) Unknown Class Discriminant Learner: UCDL includes two targets mining methods as follows.

a) Background Suspected Unknown Targets Mining: Traditional confidence score-based open-set learning methods often fail to separate unknown targets from substantial background clutter in SAR images. Based on the prior knowledge that unknown targets are generally more similar to known targets than to background regions, UCDL leverages the

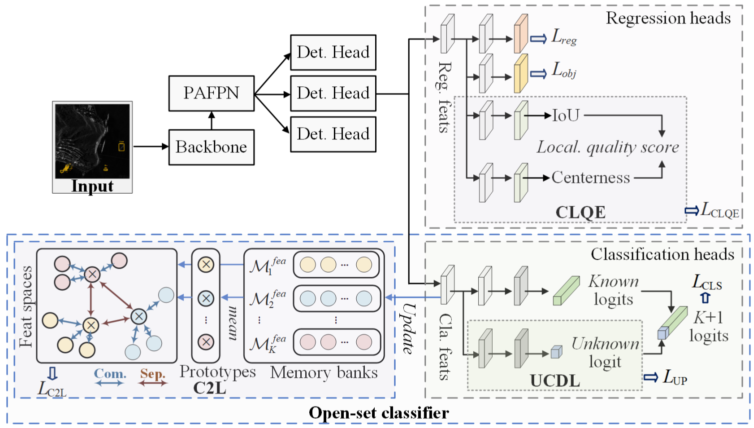


Fig. 1. Overall diagram of UTM-YOLOX.

localization quality scores of predicted boxes to preliminarily filter suspected unknown targets

$$s >= T_s \tag{3}$$

where s represents the localization quality score, T_s is the threshold for the localization quality estimation score, which follows the commonly adopted setting [11] of 0.7 in most object detection studies. Next, the top N highest-scoring targets are selected as the final unknown targets

$$\text{score} = (\text{sum}(p_c, c \in C_K \cup C_U)) \tag{4}$$

where p_c represents the logit of the cth class. This method prioritizes selecting regions with higher similarity to known targets as suspected unknown targets, effectively reducing the likelihood of mistakenly selecting meaningless background as suspected unknown targets. UCDL uses binary cross-entropy loss to constrain the scores of suspected unknown targets in the OSC as follows:

$$L_{\mathrm{UP\_UF}} = -\sum_{i=0}^{C} y_i \cdot \log(p_i) + (1 - y_i) \cdot \log(1 - p_i) \tag{5}$$

where p_i represent unknown logit, y is training label, which is 0 for known classes and 1 for unknown classes, $L_{\mathrm{UP\_UF}}$ enhances the ability to identify unknown targets and reduces the probability of misclassifying them as known classes.

b) Foreground Hard Targets Mining: Hard targets are characterized by high uncertainty and may overlap with true unknown targets within the feature space. Therefore, building upon the previous step, UCDL further selects hard foreground targets as suspected unknown targets and incorporates them into the training of OSC.

Specifically, UCDL adopts an uncertainty-guided hard targets mining method to select hard targets from foreground. Inspired by Zhao et al. [3], maximum cross-entropy is used to estimate the uncertainty of foreground targets

$$H(p) = -\sum_{c \in C_K} p_c \cdot \log(p_c) \tag{6}$$

where p_c is the probability that a sample belongs to class c.

It is important to note that these hard-to-classify foreground targets still inherently belong to the labeled known targets. However, we aim to ensure that these hard targets are correctly classified, rather than being misclassified as belonging to the unknown classes. Therefore, we control the training labels of these samples to ensure that their unknown class logit is lower than that of their true class. The training labels p_{target} can be calculated as follows:

$$p_{\text{target}} = \tau \times w \tag{7}$$

$$w = \max(p_c, c \in C_K) \tag{8}$$

where p_c represents the logit of the cth known class, and τ is hyperparameter which is set to 0.4 in our experiments. Finally, the hard targets are trained using cross-entropy loss, with the loss function defined as follows:

$$L_{\mathrm{UP\_FG}} = -\sum_{i=0}^{top-k} (p^i_{\text{target}} \cdot \log(p^i_u) + (1 - p^i_{\text{target}}) \cdot \log(1 - p^i_u)) \tag{9}$$

where p^i_u represents the unknown class logit. Equations (5) and (9) together constitute the training loss L_{UP}.

2) Contrastive Clustering Learning: The attitude sensitivity of radar targets' echoes results in poor feature separability between different target classes [6], which make it challenging for the OSC to accurately distinguish between known and unknown targets. To this end, C2L enhances the intra-class compactness of known targets to indirectly improve the model's ability to separate different target classes in such complex scenarios. To apply C2L within YOLOX, this letter maintains a dynamic class-level queue in the classification heads to store feature vectors f_c of known targets, with the mean serving as class prototype. The loss function is as follows:

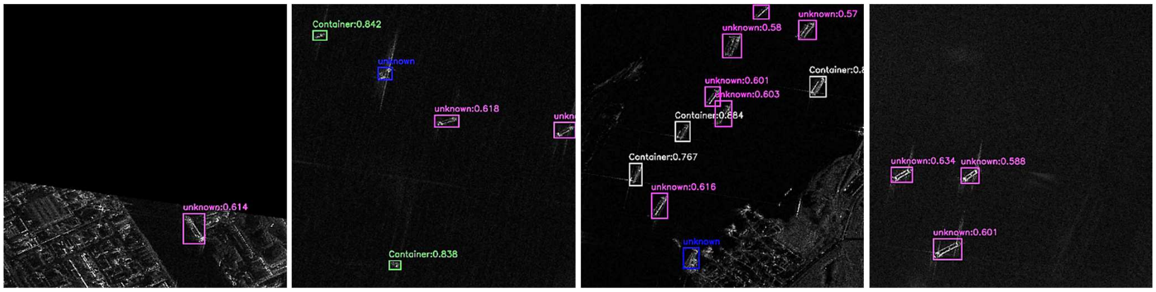
$$L_{\mathrm{CL}} = \sum_{i}^{C_K} \psi(f_c, p_i) \tag{10}$$

$$\psi(f_c, p_i) = \begin{cases} D(f_c, p_i), & i = c \\ \max\{0, \Delta - D(f_c, p_i)\}, & \text{otherwise} \end{cases} \tag{11}$$

where p represents the prototype, $D(\cdot)$ represents Euclidean distance, Δ determines the allowable proximity between similar and dissimilar items. The overall training loss is formulated

$$L_{\text{total}} = \alpha L_{\text{reg}} + \beta L_{\text{cls}} + \delta(L_{\text{obj}} + L_{\mathrm{CLQE}}) + \varepsilon(L_{\mathrm{UP}} + L_{\mathrm{CL}}) \tag{12}$$

where α, β, δ, and ε denote different hyperparameters.



(a) (b) (c) (d)

Fig. 2. Visualization of detection and classification results. (Green: known targets are correctly detected and classified; Purple: unknown targets are correctly detected and identified; Blue: missing detections; Red: false alarms; White: targets are correctly detected but wrongly classified). (a) Test image 1. (b) Test image 2. (c) Test image 3. (d) Test image 4.

E. Open-Set Test Stage

UTM-YOLOX multiplies the localization quality score and the unknown class logit as the final unknown class probability. For each predicted box, the classification heads will output a vector of $K+1$ dimensions

$$p_{\text{pred}} = \max(o \times p_1, \ldots, o \times p_k, s \times p_u) \tag{13}$$

where o represents the objectness score, s represents the localization quality score, and p_i represents the logit for ith class.

III. Experiment Results and Analysis

A. Dataset

We use the multiclass SAR ship detection and classification dataset SRSDD-v1.0 [10] to validate the open-set performance of UTM-YOLOX, which contains six types of ship targets. This letter converts the original rotated boxes into horizontal boxes using the method of maximum bounding rectangle.

B. Experiment Settings

SRSDD-v1.0 contains six classes of ship targets, of which container, ore-oil, fishing, law-enforce, and cell-container are set as known classes, and Dredger as unknown classes. In the training stage, according to the $\text{OOD}_{\text{unlabeled}}$ assumption, we only use the known class labels to train the detector. In the testing stage, both known and unknown targets are used to evaluate the open-set performance. According to the original YOLOX [7], α, β, and δ are set to 5.5, 1, and 1, respectively. Since the L_{cls} and the $(L_{\text{UP}} + L_{\text{CL}})$ are focus on the classification performance of known classes and unknown targets, respectively, the size of ε determines whether the UTM-YOLOX focuses on the known classes and unknown targets. In our experiment, we expect satisfactory classification performance for both known classes and unknown targets, so that the ε is set to 0.5.

C. Evaluation Metrics

We use seven metrics to evaluate the open-set performance, including precision, recall, $f1$-score, unknown detection rate (UDR), and unknown detection precision (UDP) [11]

$$\text{Precision} = \frac{\text{TP}}{\text{TP+FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP+FN}} \tag{14}$$

$$F1-\text{score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{15}$$

$$\text{UDR} = \frac{\text{TP}_u\text{+FN}_u^*}{\text{TP}_u\text{+FN}_u}, \quad \text{UDP}=\frac{\text{TP}_u}{\text{TP}_u\text{+FN}_u^*} \tag{16}$$

where TP is true positives, FP is false positives, and FN is false negatives. TP_u is the number of unknown targets that are correctly detected and identified. FN_u^* is the number of unknown targets that are correctly detected but incorrectly identified. FN_u is the number of missing detections and unknown targets detected correctly but misclassified. UDR evaluates the ability to detect unknown targets and UDP evaluates the ability to identify unknown targets [3]. KE is the number of known targets incorrectly identified as unknown classes. UFA is the background identified as unknown targets.

D. Open-Set Experiments

Our method is compared with three recent open-set detection and classification methods, including Random Box [3], PAD-CEC [4], and OW-DETR [12]. As shown in Table I, UTM-YOLOX achieves the best overall open-set performance. By reliably mining unknown targets for training the OSC, UTM-YOLOX effectively mitigates confusion among known targets, unknown targets, and background clutter, leading to superior metrics on known, unknown, and average. In contrast, the other methods are not specifically designed for SAR target open-set detection and classification tasks and struggle with foreground–background confusion and low interclass separability, resulting in suboptimal performance. Compared to our method, Random Box achieves higher known-target recall and UDR at the cost of substantial computational overhead, as it randomly generates a large number of candidate boxes on the images. However, a pseudo-label selection strategy that relies solely on the original confidence tends to misclassify strong clutter backgrounds as potential unknown targets, leading to a decrease in precision. Therefore, the $F1$ score of Random Box is lower than that of our method. Fig. 2 shows the visualization results of UTM-YOLOX on four test images. We can see that our method effectively detects and identifies unknown targets while accurately detecting and classifying known targets.

E. Ablation Experiment

This section evaluates the contributions of the three modules in UTM-YOLOX—CLQE, UCDL, and C2L—to open-set performance through ablation experiments. The results are presented in Table II. The model without any module cannot classify unknown targets (precision = 0), resulting in an $F1$-score of 0 of unknown targets. The model with CLQE

TABLE I
OPEN-SET PERFORMANCE OF DIFFERENT METHODS (%)

Methods	class	Precision ↑	Recall ↑	F1-score ↑	UDR ↑	UDP ↑
Random Box[2]	Known	76.3	**72.0**	72.2	-	-
	unknown	21.2	15.2	17.7	**78.3**	19.4
	Average	67.1	62.5	63.2	-	-
PAD-CEC[3]	Known	68.3	66.2	64.1	-	-
	unknown	30.0	26.1	27.9	65.2	40.0
	Average	61.9	59.5	58.1	-	-
OW-DETR[12]	Known	70.4	64.9	67.5	-	-
	unknown	26.6	33.5	29.7	64.0	41.3
	Average	63.1	59.6	61.3	-	-
UTM-YOLOX	Known	**85.8**	69.2	**74.8**	-	-
	unknown	**48.7**	**41.3**	**44.7**	71.7	**57.6**
	Average	**79.7**	**64.5**	**69.8**	-	-

TABLE II
ABLATION EXPERIMENTS (%)

CLQE	UCDL	C2L	F1-score K.	F1-score U.	UDR	UDP
×	×	×	70.3	0.0	45.7	0.0
√	×	×	53.4	17.7	84.8	94.9
√	√	×	72.7	38.6	71.7	42.4
√	√	√	**74.8**	**44.7**	**71.7**	**57.6**

TABLE III
INFERENCE SPEED (SECOND)

Methods	Random Box	PAD-CEC	OW-DETR	UTM-YOLOX
Inference time	549.05	51.84	18.61	8.73

uses the localization quality estimation score as the classification probability for unknown classes, enabling accurate detection, and classification of unknown targets, with UDR and UDP reaching 84.8% and 94.9%, respectively. However, due to strong clutter in SAR images and small interclass differences among target categories, the model produces numerous false positives (leading to reduced precision) and misclassifications (leading to reduced recall). As a result, the $F1$-scores of known and unknown targets remain low. UCDL effectively selects high-quality suspected unknown targets from background clutter to train the OSC, significantly reducing false positives and misclassification of unknown targets. With the introduction of C2L, the separability between different targets is enhanced, resulting in $F1$-score improvements of 3.1% for known targets and 6.1% for unknown targets.

F. Inference Speed

The experimental results of the inference time on an NVIDIA RTX 3080 for all methods are shown in Table III. Benefiting from adopting the single-stage YOLOX detector [8] as the backbone and designing a simple yet efficient open-set learning strategy, the proposed UTM-YOLOX achieves the best inference time compared to other methods, with an inference time of 8.73 s, which is only half that of OW-DETR.

IV. CONCLUSION

This letter proposes an open-set detector for SAR ship detection and classification, achieving satisfactory performance on the measured dataset. However, the method relies on the presence of unlabeled unknown targets during training and cannot distinguish between different unknown targets. Future work will explore scenarios without unknown targets to further improve generalization and develop the ability to differentiate diverse unknown classes for enhanced interpretability.

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